Analysis and Prediction of Factors influencing Nuclear Power Plant Cost

DATA 603: Statistical Modelling with Data

Project Report

Group Number : 10

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Introduction

- Historically, nuclear power has always been quite an underutilised source for electricity generation. One of the major reasons why nuclear fuels like Uranium, Plutonium and Thorium were not able to displace Coal was because Nuclear Power plant are too costly to construct.
- Our topic is about analysing numerous factors which contribute to the expensiveness of those reactors. For us, this topic is important because Nuclear is an Eco-friendly source of power generation and should be widely adopted over Coal Power.
- Our end goal is to identify and discover the most statistically significant predictor variables while keeping "Costs ($)" as dependent variable. Finally, we want to find the best fit model which can predict the cost based on predictor variables and various interaction terms.
- Through this, we can then know on which factors should humanity work on to make Nuclear Reactors as cheap as possible.

Dataset

The main time consuming task for our project was Data Sourcing and Data Cleansing our csv file. Around 80-90% of the effort went into collating the final csv file.

Sources from where the dataset was collated

Our final CSV file was made with the combination of data taken from the following websites linked below: -

1) *Construction Time*, *Start year*, *Country*, *Capacity*, *Reactor Type* and *Frequency* variables were sourced from : - https://pris.iaea.org/PRIS/CountryStatistics/CountryDetails.aspx?current=IN
2) *Steel cost* was sourced from : - https://fred.stlouisfed.org/series/WPU101
3) *Cement cost* was sourced from: - https://fred.stlouisfed.org/series/WPU1333
4) *Inflation Rate* was sourced from : - https://data.worldbank.org/indicator/FP.CPI.TOTL.ZG
5) *Cost of reactors* were sourced from : -

- *AP1000's:* https://de.nucleopedia.org/wiki/Advanced_Passive#Auftr%C3%A4ge
- *VVER-1200:* https://de.nucleopedia.org/wiki/WWER-1200#Auftr%C3%A4ge
- *VVER-1000:* https://de.nucleopedia.org/wiki/Liste_der_WWER#Version_392
- *EPR:* https://de.nucleopedia.org/wiki/Framatome_EPR#Auftr%C3%A4ge
- *CNP-300:* https://de.nucleopedia.org/wiki/Benutzer:TZV/CNP-300#Auftr%C3%A4ge

- *Hualong One:*https://de.nucleopedia.org/wiki/Benutzer:TZV/HPR#Auftr%C3%A4ge

Note: - Most of the data we sourced is from the official **IAEA** website. IAEA is a UN Body which stands for **International Atomic Energy Agency.**

Sampling of the dataset: -

Nuclear_Data

Cost_Billion_Dollars	Name	Country	Reactor_Type	Gross_Capacity_MWe	Thermal_Capacity_MWt	Efficiency_Percentage	Starting_Year	Duration	Average_Inflation_Percentage	Average_Concrete_Price	Average_Steel_Price
1.261	ANO-1	UNITED STATES OF AMERICA	PWR	903	2568	35.16	1968	6	0.0488	37.123333	34.33333333

Table : 1 *The table above shows a small sample of our final CSV file.*

Explanation of the first row of our dataset: -

- **1.261** is the cost of the nuclear reactor in Billion US $ (year 2007).
- **ANO-1** is the name of the said reactor.
- **UNITED STATES OF AMERICA** is the name of the country where reactor was constructed.
- **PWR** is the nuclear reactor type. PWR stands for "Pressurised Water Reactor".
- **903** is the amount of electricity that reactor generates (in Mega Watts electric).
- **2568** is the amount of heat that nuclear reactor generates (in Mega Watts Thermal).
- **35.16** is the percentage amount. How much % is *ANO-1* reactor able to *convert* from **heat into electricity**. (It basically shows how **efficient** is the Turbine which converts heat to electricity.)
- **1968** is the year when it started getting constructed.
- **6** signifies that it took *6 years* for the reactor to get completed. I.e. from 1968 to 1974, the reactor was getting built.
- **0.0488** shows the Average Year-on-Year (YoY) inflation percentage from the year 1968 to 1974.
- **37.12333** shows the average Concrete price from the year 1968 to 1974.
- **34.33333** shows the average Steel price from the year 1968 to 1974.

Methodology

Below is the list given of our Predictor variables, which we have used in our project: -

S.No.	Parameters	Qualitative/ Quantitative Variable	Units of each parameter along with the details to derive the value if needed.
1	Cost of Power plant (Dependent Variable)	Quantitative (Continuous)	**Cost (Billion US $)** (Year 2007)
2	Duration (Predictor Variable)	Quantitative (Continuous)	**Years** (Total number of years it took to build the reactor)
3	Construction Start Year (Predictor Variable)	Quantitative (Discrete)	The year in which the construction of the reactor **started**.
4	Inflation (Predictor Variable)	Quantitative (Continuous)	Inflation % (Average Year-on-Year (**YoY**) inflation percentage during the duration of reactor construction)
5	Electrical Capacity (Predictor Variable)	Quantitative (Continuous)	Net Electrical Capacity to generate power (**MWe**)
6	Thermal Capacity (Predictor Variable)	Quantitative (Continuous)	Gross Thermal Capacity of the reactor vessel (**MWt**)
7	Country (Predictor Variable)	Qualitative (21 Countries)	Country Name
8	Reactor Type (Predictor Variable)	Qualitative (7 types)	Reactor Type Name
9	Efficiency of Power Plant (Predictor Variable)	Quantitative (Continuous)	(Net Capacity / Thermal Capacity)*100 (%)
10	Concrete Price (Predictor Variable)	Quantitative (Continuous)	Average price of concrete during construction time (**PPI**)
11	Steel Price (Predictor Variable)	Quantitative (Continuous)	Average price of steel during construction time (**PPI)**

Table : 2 *The table above summarises the variables in our model.*

Further details of our predictor variables: -

1) *Country*: - Qualitative variable which lists the name of the country where the nuclear reactor was made. 21 types of countries are listed in our csv file.
The list of countries mentioned in our CSV file are: -
UNITED STATES OF AMERICA , RUSSIA ,FRANCE , INDIA, JAPAN, UNITED KINGDOM, SWEDEN , KOREA, REPUBLIC OF UNITED ARAB EMIRATES ,GERMANY , CZECH REPUBLIC , PAKISTAN, SWITZERLAND, SLOVAKIA, MEXICO , IRAN, ISLAMIC REPUBLIC OF ARGENTINA, BELARUS, BRAZIL , CANADA, CHINA

2) *Reactor Type*: - Nuclear energy can be implemented in many different ways; hence the different types of Nuclear Reactors. In our data set, we have a list of 7 different reactor types.

The list of Reactor mentioned in our CSV file are: -

- PWR (Pressurised Water Reactor)
- BWR (Boiling Water Reactor)
- FBR (Fast Breeder Reactor)
- HTGR (High Temperature Gas Cooled Reactor)
- LWGR (Light Water Graphite Reactor)
- PHWR (Pressurised Heavy Water Reactor)
- GCR (Gas Cooled Reactor)

3) *Gross Capacity (MWe):* - The amount of electricity the nuclear reactor can generate during its operation. The units are in Mega Watts Electric.

4) *Thermal Capacity (MWt):* - The amount of heat the nuclear reactor generates due to fission. Do note that heat is converted to electricity. The units are in Mega Watts Thermal.

5) *Efficiency Percentage:* - How efficient is the nuclear reactor in converting heat to electricity? The more the efficiency; the more heat is converted to electricity.

6) *Starting Year:* - The year when the construction of that nuclear reactor started.

7) *Duration (in years):* - The number of years it took to make that nuclear reactor from start to end.

8) *Average Inflation Percentage:* - The amount of inflation that occurred (YoY) during the time period it took for construction of the reactor.

9) *Average Concrete Price:* - The average concrete price during the time period it took for construction of the reactor. (**Producer Price Index** (**PPI)** value is given)

10) *Average Steel Price:* - The average steel price during the time period it took for construction of the reactor. (**Producer Price Index** (**PPI)** value is given)

Reasons for taking the following predictor variables: -

1) **Duration: -** We would like to know whether nuclear reactors which are made quicky are cheaper to make or whether the cost of nuclear reactor increases due to lengthy construction times.
2) **Construction Start Year: -** We would like to know whether the reactors made in early 60's or 70's were cheaper or more expensive to make compared to today.
3) **Inflation: -** We are thinking of taking inflation into account as well and want to know whether during the times of high inflation rates, does the cost of nuclear reactor increases/decreases or have no effect?
4) **Net Capacity: -** A nuclear power plant consists of 2 parts. First is "Nuclear Island" and the second part is "Conventional Island." A nuclear power plant having a greater capacity to generate electricity should be costlier or cheaper, that is to be found out.
5) **Thermal Capacity: -** The greater the amount of heat a reactor can generate in an hour usually means that the reactor vessel is huge in size and capacity. Does having bigger reactor vessels result in costlier reactors?
6) **Country: -** We have anecdotally heard that countries like India, China and Russia are just superior when it comes to cheap reactor construction. Is this true? Does some countries make costlier reactors compared to other countries?
7) **Reactor Type: -** Electricity through nuclear can be produced through various ways i.e. different reactor types. Do some reactor types more cheaper than the other types?
8) **Efficiency of Power Plant: -** The more efficient a power plant is, the bigger and costlier the generator is in a reactor. Does the cost of a better generator is significant enough to increase the cost of the reactor as a whole?
9) **Concrete Price: -** Majority of the construction material used in a nuclear reactor is made up of concrete and steel. Does higher concrete prices in a specific time-period results in higher cost of construction?
10) **Steel Price: -** Majority of the construction material used in a nuclear reactor is made up of concrete and steel. Does higher steel prices in a specific time-period results in higher cost of construction?

Hypothesis

Our Hypothesis of how the predictor variables should affect the response variable are as follows: -

1) *Country:* - We expect to see different prices of nuclear reactors on the basis of different countries due to various differences in labour costs, laws and regulations, financing methods which are particular for that specific country etc.
2) *Reactor Type:* - We expect to see that the reactor type "PWR" be the cheapest reactor to be constructed because it is the most common type of nuclear reactor to be made. Economies of scale dictates that enterprises obtain due to their scale of operation more cost advantages over others who cannot scale.

   On a similar note, we expect to see "FBR" reactor to be the most costly one because it is one of the most challenging reactor to be made.

3) *Gross Capacity (MWe):* - The more electricity a nuclear reactor generates, the more costly the reactors should be, since the conventional island would be bigger in size.
4) *Thermal Capacity (MWt):* - The more heat a reactor generates, the more costly that reactor should be, since the reactor core would be bigger in size.
5) *Efficiency Percentage:* - The more efficient the reactor is, it means the more expensive turbine and generator in the conventional island was used. Hence more the efficiency, more the cost.
6) *Starting Year:* - We expect to see the reactors which were made in earlier years to be cheaper than the reactors which are made in modern times due to less safety regulations, lax laws and ease of financing during earlier times.
7) *Duration:* - The more time a reactor takes to be constructed, the more costly it should be due to cost overruns etc.
8) *Average Inflation Percentage:* - The more the inflation is during a certain period of time, the costlier the things would be.
9) *Average Concrete Price:* - Over 95% of materials which makes a nuclear reactor is steel and concrete. We expect to see, the more the average price of steel, the more the reactor cost.
10) *Average Steel Price:* - Over 95% of materials which makes a nuclear reactor is steel and concrete. We expect to see, the more the average price of steel, the more the reactor cost.

Statistical Analysis Techniques

We would be using Multiple Linear Regression modelling on our dataset and find out the significant predictor variables which affects our response variable the most.

Workflow Diagram of our Project

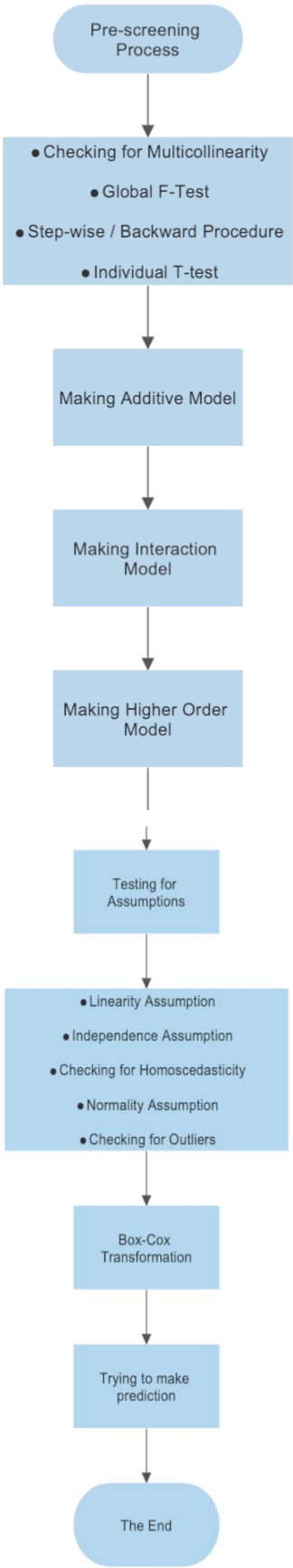

Figure : 1(*The figure above shows the whole workflow process we adopted in order to complete our project.*)

Note: - For all the above tests, we have taken α = 0.05.

Reason for following the workflow process mentioned above: -

1) *Checking for Multicollinearity:* - We have to check for Multicollinearity first and foremost in order to prevent coefficient estimates to swing wildly based on which other independent variables are in the model. We have to eliminate Multicollinear variables because these variables reduces the precision of the estimate coefficients, which weakens the statistical power of our regression model.
2) *Global F-Test:* - This test is being performed in order to ascertain that there is at least 1 predictor variable in our dataset which is significant enough for our response variable.
3) *Stepwise / Backward Procedure:* - We are doing these procedures in order to know the most significant and least significant predictor variable in our dataset. Also we would like to know which predictor variables are significant enough for our response variable.
4) *Individual T-Test:* - We are performing Individual T-test in order to get our first order additive model
5) *Making Additive Model:* - We would be making our Additive model and then also perform Partial F-Test in order to confirm that the predictor variables which we excluded have been a correct decision or not.
6) *Making Interaction Model:* - After our Additive Model, we would be looking for any interaction terms within them in order to make our model more accurate.
7) *Making Higher Order Model:* - After our interaction model, we would try to make a higher order model in order to improve the $R^2 Adjusted$ of our Multiple Linear Regression Model.
8) *Testing for Assumptions:* - After getting our best model, we shall be testing for all the assumptions in order to make our Multiple Linear Regression Model work.
9) *Box-Cox Transformation:* - We tried to do Box-Cox transformation because one of our assumptions failed.
10) *Making Prediction:* - We tried to make a prediction from our model but it was not successful. We shall explain the reason in detail in the result section.

Workload Distribution

Our dataset was not readily available. We collected data and had to manually combine the parameters into a single CSV.

Data collection in our case was the most time consuming task. Devanshi developed script for web scraping (Figure 2) whereas Wan and Jelilat helped to find the manual data part for the datasets. The Cost (which is our response variable) was critical and needed to be precise thus Swarnim invested all his time in collecting the data for that column.

```
C: > Users > Devanshi > WebScrapping > ND.py > ...
elements_India = ['289','290','291','292','308','286','986','852','853','1059','304','305','306','307','302','303','287','288','295','297','285','296','300','301']
elements_Japan = ['345','377','389','334','340','341','342','350','351','361','364','336','352','372','373','335','349','362','376','843','842','348','358','374','333','359','365','380','381','367','383','384','366','386','338','357','339','343','
elements_Ukraine = ['568','569','575','576','573','574','570','571','572','589','577','578','579','584','586','594','595','566','567']
elements_UK = ['264','265','266','267','231','232','233','234','252','253','254','255','242','243','275','276','248','249','250','251','256','257','259','260','271','272','244','245','269','270','246','247','237','238','235','236','263','261','262
elements_Spain = ['153','154','155','156','140','139','149','141','150','146']
elements_Swedan = ['528','538','540','541','530','532','534','535','531','536','537','539','529']
elements_Koreo = ['411','412','396','397','402','403','413','395','398','399','404','406','394','405','409','410','885','886','887','864','865','883','884','407','408','400','401']
elements_Finland = ['157','158','159','160','161']
elements_UAE = ['1050','1051','1052']
elements_Germany = ['114','85','91','107','86','108','97','123','124','125','126','127','101','104','100','102','89','106','135','94','133','96','93','88','119','121','87','98','122','83','92','90','136']
elements_Czech = ['78','79','80','81','74','75']
elements_Pakistan = ['428','429','1044','1045','427','1067','1068']

elements_Swizerland= ['55','57','58','59','966','56']
elements_Hungary = ['278','279','280','281']
elements_Slovakia = ['543','548','549','546','547','550','551']
elements_SouthAfrica = ['836','837']
elements_Mexico = ['420','421']
elements_Romania = ['442','443']
elements_Iran = ['310']
elements_Slovenia = ['542']
elements_Netherlands =['423','422']
elements_Lithunia = ['415','416']
elements_Italy = ['327','326','325','318']
elements_Kazakhstan = ['414']

Country_lst= [elemts_USA,elemts_Russia,elemts_France,elemts_India,elemts_Japan,elemts_Ukraine,elemts_UK,elemts_Spain,elemts_Swedan,elemts_Koreo,elemts_Finland,elemts_UAE,elemts_Germany,elemts_Czech,elemts_Pakistan,elemts_Swizerland,elemts_Hungar
#Country_lst= [elemts_UAE]
df = pd.DataFrame(columns = ['Name','Country','Reactor_Type','Gross_Capacity_MWe','Thermal_Capacity_MWt','Efficiency_Percentage','Starting_Year','End_Year','Duration','Inflation'])
for every_ele in Country_lst:
    for ele in every_ele:
        driver.get(URl+ele)
        print(ele)


        #for number in range(1,2):
        Name = driver.find_elements("xpath",'/html/body/form/div[3]/div[3]/div/div[1]/div/h1/span[1]')
        Country = driver.find_elements("xpath",'/html/body/form/div[3]/div[3]/div/div[2]/div/ul/li[1]/a')
        Reactor_Type = driver.find_elements("xpath",'/html/body/form/div[3]/div[3]/div/div[1]/div/div[2]/div[2]/table/tbody/tr[2]/td[1]/h5/span')
        Gross_Capacity_MWe = driver.find_elements("xpath",'/html/body/form/div[3]/div[3]/div/div[1]/div/div[2]/div[2]/table/tbody/tr[4]/td[3]/h5/span')
        Thermal_Capacity_MWt = driver.find_elements("xpath",'/html/body/form/div[3]/div[3]/div/div[1]/div/div[2]/div[2]/table/tbody/tr[4]/td[4]/h5/span')
        Efficiency_Percentage = driver.find_elements("xpath",'/html/body/form/div[3]/div[3]/div/div[1]/div/div[2]/div[2]/table/tbody/tr[4]/td[4]/h5/span')
        Starting_Year = driver.find_elements("xpath",'/html/body/form/div[3]/div[3]/div/div[1]/div/div[2]/div[2]/table/tbody/tr[6]/td[1]/h5/span')
        End_Year = driver.find_elements("xpath",'/html/body/form/div[3]/div[3]/div/div[1]/div/div[2]/div[2]/table/tbody/tr[6]/td[2]/h5/span')
        Duration = driver.find_elements("xpath",'/html/body/form/div[3]/div[3]/div/div[1]/div/div[2]/div[2]/table/tbody/tr[6]/td[1]/h5/span')

        Upper = int(Gross_Capacity_MWe[0].text)
        Lower = int(Thermal_Capacity_MWt[0].text)

        str1 = str(Starting_Year[0].text)
        str2 = str(End_Year[0].text)

        Start = int(str[:4:12])
```

Figure : 2 *The code written for Web Scraping in order to get data from IAEA website.*

For the decision on which tests and which statistical method to use for MLR, the whole group sat together to discuss and implement the code for the same.
Once the code was completed, Devanshi and Swarnim were responsible for the Project Report creation while Wan and Jelilat reviewed and pointed out any discrepancy found in the document.
Each and every member of our group then sat together in order to make the PPT.
We can confidently say that each and every member's workload has been equal.

Results

Multicollinearity

We calculate VIF for our model with all the predictor variables.

Our model equation is given below: -

$$multicollinearity.nuclear.regression.model < -lm(Cost_Billion_Dollars \sim Gross_Capacity_MWe + Thermal_Capacity_MWt + Efficiency_Percentage + Starting_Year + Duration + Average_Inflation_Percentage + Average_Concrete_Price + Average_Steel_Price, data = nuclear.df)$$

```
 VIF Multicollinearity Diagnostics

                                   VIF detection
Gross_Capacity_MWe           126.0471         1
Thermal_Capacity_MWt         114.5285         1
Efficiency_Percentage          2.1803         0
Starting_Year                 48.0230         1
Duration                       1.7462         0
Average_Inflation_Percentage   1.0866         0
Average_Concrete_Price        78.4207         1
Average_Steel_Price           52.4073         1

Multicollinearity may be due to Gross_Capacity_MWe Thermal_Capacity_MWt Starting_Year
Average_Concrete_Price Average_Steel_Price regressors

1 --> COLLINEARITY is detected by the test
0 --> COLLINEARITY is not detected by the test
```

Figure : 3 *The result of our VIF*

As we can see, Multicollinearity is detected between $Gross_Capacity_Mwe$, $Thermal_Capacity_MWt$, $Starting\ Year$, $Average_Concrete_Price$ and $Average_Steel_Price$

We have detected collinearity between several variables, We must first concentrate on those 2 variables with the highest VIF value, i.e. Gross_Capacity_MWe and Thermal_Capacity_MWt and remove one of them.

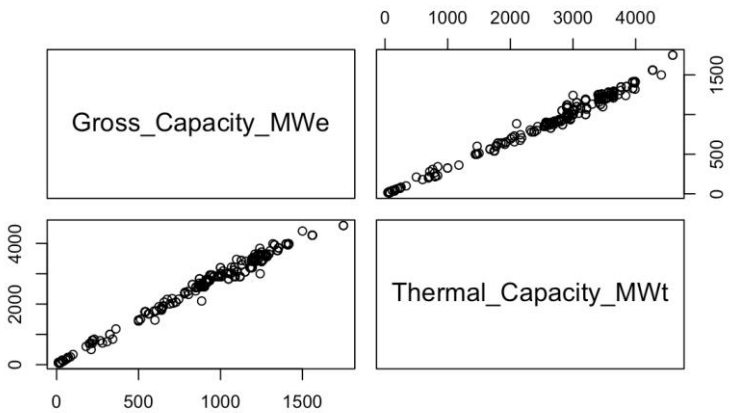

Figure : 4 *Scatter Plot*

Since, VIF value of $Gross_Capacity_Mwe > Thermal_Capacity_MWt$. Therefore we removed $Thermal_Capacity_MWt$ from our model.

Checking Multicollinearity once again after removing $Thermal_Capacity_MWt$: -

```
 VIF Multicollinearity Diagnostics

                                VIF detection
Gross_Capacity_MWe           1.6770         0
Efficiency_Percentage        1.6309         0
Starting_Year               46.7577         1
Duration                     1.6575         0
Average_Inflation_Percentage 1.0254         0
Average_Concrete_Price      77.8593         1
Average_Steel_Price         51.4576         1

Multicollinearity may be due to Starting_Year Average_Concrete_Price
Average_Steel_Price regressors

1 --> COLLINEARITY is detected by the test
0 --> COLLINEARITY is not detected by the test
```

Figure : 5 *The result of our VIF (2nd time)*

As we can see, Multicollinearity is detected between $Starting\ Year$, $Average_Concrete_Price$ and $Average_Steel_Price$.

We have detected collinearity between several variables, We must first concentrate on those 2 variables with the highest VIF value, i.e. $Average_Concrete_Price$ and $Average_Steel_Price$ and remove one of them

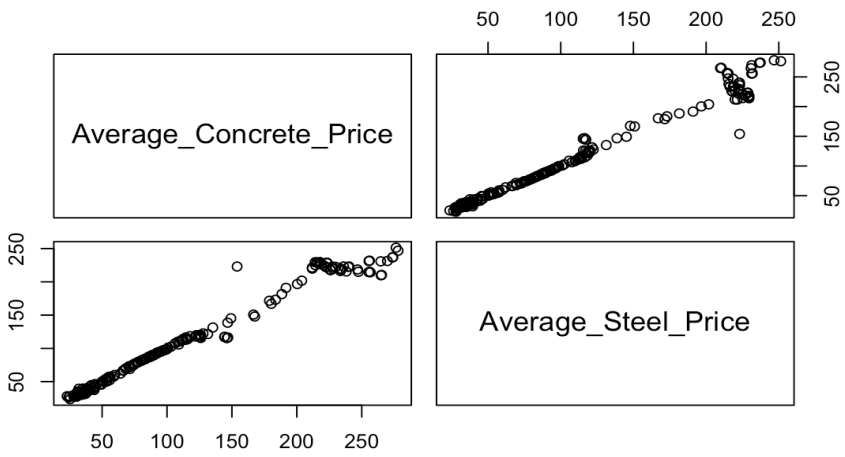

Figure : 6 *Scatter Plot*

Since, VIF value of $Average_Concrete_Price > Average_Steel_Price$. Therefore, we removed $Average_Steel_Price$ from our model.

Checking Multicollinearity once again after removing $Average_Steel_Price$: -

```
 VIF Multicollinearity Diagnostics

                                   VIF detection
Gross_Capacity_MWe              1.6767         0
Efficiency_Percentage           1.6247         0
Starting_Year                  44.9093         1
Duration                        1.6144         0
Average_Inflation_Percentage    1.0214         0
Average_Concrete_Price         45.1784         1

Multicollinearity may be due to Starting_Year Average_Concrete_Price regressors

1 --> COLLINEARITY is detected by the test
0 --> COLLINEARITY is not detected by the test
```

Figure : 7*The result of our VIF (3rd time)*

As we can see, Multicollinearity is detected between $Starting\ Year\ and$ $Average_Concrete_Price$.

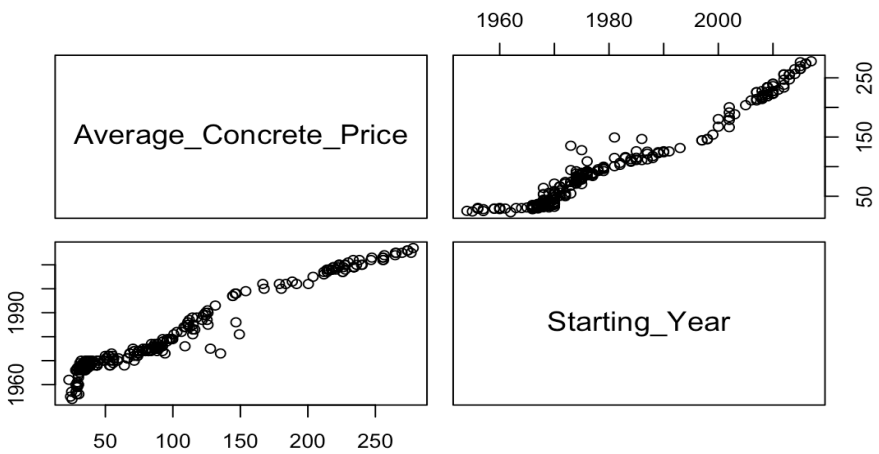

Figure : 8 *Scatter Plot*

Since, VIF value of $Average_Concrete_Price > Starting\ Year$. Therefore we removed $Starting\ Year$ from our model.

Checking Multicollinearity once again after removing $Starting\ Year$: -

```
 VIF Multicollinearity Diagnostics

                                VIF detection
Gross_Capacity_MWe           1.6346         0
Efficiency_Percentage        1.6150         0
Duration                     1.0928         0
Average_Inflation_Percentage 1.0212         0
Average_Concrete_Price       1.1643         0

NOTE:  VIF Method Failed to detect multicollinearity


0 --> COLLINEARITY is not detected by the test
```

Figure : 9

The result of our VIF (4th time)

No more multicollinearity found between the variables. Our new equation is:

$$new.nuclear.regression.model < -lm(Cost_Billion_Dollars \sim factor(Country) + factor(Reactor_Type) + Gross_Capacity_MWe + Efficiency_Percentage + Duration + Average_Inflation_Percentage + Average_Concrete_Price, data = nuclear.df)$$

Global F-Test

We ask the global question, Is this multiple regression model any good at all? The answer is that we can test some hypotheses to see the relationship between the response and predictors. The first of these hypotheses is an overall F-test or a global F test which tells us if the multiple regression model is useful. To address the overall question, we will test

$$H0: \beta 1 = \beta 2 = \ldots = \beta 7 = 0$$

$$Ha: At\ least\ one\ \beta i\ is\ not\ zero\ (i = 1,2, \ldots ,7)$$

Our null hypothesis basically means that all of our supposedly Independent variables are junk food and not useful at all.

Our Alternative Hypothesis means that at least one of the independent variable is useful in our model.

We therefore shall now do ANOVA testing.

```
Analysis of Variance Table

Model 1: Cost_Billion_Dollars ~ 1
Model 2: Cost_Billion_Dollars ~ factor(Country) + factor(Reactor_Type) +
    Gross_Capacity_MWe + Efficiency_Percentage + Duration +
Average_Inflation_Percentage +
    Average_Concrete_Price
  Res.Df    RSS Df Sum of Sq      F                Pr(>F)
1    260 1425.9
2    229  660.8 31    765.14 8.5535 < 0.00000000000000022 ***
```

Figure : 10 *The result of our Global F-Test*

Since the p − value of the ANOVA table is <0.05, it means that the null hypotheses has to be rejected. This means that at least one independent variable in our model is NOT junk.

Step-wise Regression Procedure

We shall now do Step-wise Regression Procedure in order to know the significant variables in our regression model.

$$P_{ent} = 0.1$$

$$P_{rem} = 0.3$$

```
                                          Pr(>|t|)
(Intercept)                               0.039787 *
Gross_Capacity_MWe                      0.00000684 ***
Duration                                0.00004660 ***
factor(Reactor_Type)FBR                   0.000355 ***
factor(Reactor_Type)HTGR                0.00003420 ***
factor(Reactor_Type)LWGR                  0.003693 **
factor(Country)BELARUS                    0.002864 **
factor(Country)IRAN, ISLAMIC REPUBLIC OF   0.014372 *
Average_Concrete_Price                  0.00001139 ***

Residual standard error: 1.691 on 231 degrees of freedom
Multiple R-squared:  0.5366,    Adjusted R-squared:  0.4784
F-statistic: 9.223 on 29 and 231 DF,  p-value: < 0.00000000000000022
```

Figure : 11 *The result of our Step-Wise Procedure*

According to Step-wise procedure, the most significant variable in our model is Gross_Capacity_MWe variable and the significant variables are $Gross_Capacity_MWe$,

$Duration$, $Average_Concrete_Price$, $Reactor_Type$ and $Country$. We shall now do Backward Regression Procedure in order to know the least significant variable in our dataset.

Backward Procedure

We shall now do Backward Regression Procedure in order to know the significant variables in our regression model.

$$P_{rem} = 0.3$$

```
                                          Pr(>|t|)
(Intercept)                               0.039787 *
factor(Country)BELARUS                    0.002864 **
factor(Country)IRAN, ISLAMIC REPUBLIC OF   0.014372 *
factor(Reactor_Type)FBR                   0.000355 ***
factor(Reactor_Type)HTGR                 0.00003420 ***
factor(Reactor_Type)LWGR                  0.003693 **
Gross_Capacity_MWe                     0.00000684 ***
Duration                               0.00004660 ***
Average_Concrete_Price                 0.00001139 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1.691 on 231 degrees of freedom
Multiple R-squared:  0.5366,    Adjusted R-squared:  0.4784
F-statistic: 9.223 on 29 and 231 DF,  p-value: < 0.00000000000000022
```

Figure : 12 *The result of our Backward Procedure*

According to backward procedure, the most insignificant variable in our model was Efficiency_Percentage variable.

The common significant variables, both in Step-wise and Backward procedure are: - are $Gross_Capacity_MWe$, $Duration$, $Average_Concrete_Price$, $Reactor_Type$ and $Country$. We shall now do Individual Coefficient Test (T-test) in order to finally know our significant variables and make our first order model.

Individual T-test

We shall now use Individual Coefficients Test (t-test) to find out the best model for our use.

$$H0 : \beta i = 0$$

$$Ha: \beta i \neq 0 \ (i = 1,2,\ldots,7)$$

```
factor(Country)BELARUS                    **

factor(Country)IRAN, ISLAMIC REPUBLIC OF *

factor(Reactor_Type)FBR                   **

factor(Reactor_Type)HTGR                  ***
factor(Reactor_Type)LWGR                  *

Gross_Capacity_MWe                        ***

Duration                                  ***

Average_Concrete_Price                    ***

Residual standard error: 1.699 on 229 degrees of freedom
Multiple R-squared:  0.5366,    Adjusted R-squared:  0.4739
F-statistic: 8.553 on 31 and 229 DF,  p-value: < 0.00000000000000022
```

Figure : 13 *The result of our Individual T-test*

From our Individual T-test, we can see that $Gross_Capacity_MWe$, $Duration$, $Average_Concrete_Price$, $Reactor_Type$ and $Country$ are the **only significant variables** in our regression model, as confirmed by Step-Wise and Backward procedure.

Efficiency_Percentage and Average_Inflation_Percentage variables are **insignificant variables**, therefore we shall remove them.

Additive Model

Our new regression model is:

$$new.nuclear.regression.model.reduced$$
$$< -lm(Cost_Billion_Dollars \sim factor(Country)$$
$$+ factor(Reactor_Type) + Gross_Capacity_MWe + Duration$$
$$+ Average_Concrete_Price, data = nuclear.df)$$

Partial F-Test

We shall now do Partial F-Test in order to confirm whether Efficiency_Percentage and Average_Inflation_Percentage variables are truly **insignificant variables or not.**

$$Ho: < List\ of\ insignificant\ variables > = 0$$

```
Analysis of Variance Table

Model 1: Cost_Billion_Dollars ~ factor(Country) + factor(Reactor_Type) +
    Gross_Capacity_MWe + Duration + Average_Concrete_Price
Model 2: Cost_Billion_Dollars ~ factor(Country) + factor(Reactor_Type) +
    Gross_Capacity_MWe + Efficiency_Percentage + Duration +
Average_Inflation_Percentage +
    Average_Concrete_Price
  Res.Df   RSS Df Sum of Sq      F Pr(>F)
1    231 660.8
2    229 660.8  2 0.0017806 0.0003 0.9997
```

Figure : 14 *The result of our Partial F-test*

The Partial F test clearly states that we cannot reject the NULL hypothesis, therefore it was a correct decision to remove all junk predictor variables.

Interaction Model

We shall now make our interaction model using the equation: -

$$interact.new.nuclear.regression.model.reduced$$
$$< -lm(Cost_Billion_Dollars \sim (factor(Country)$$
$$+ factor(Reactor_Type) + Gross_Capacity_MWe + Duration$$
$$+ Average_Concrete_Price)\text{^}2, data\ =\ nuclear.df)$$

We would be using Individual Coefficients Test (t-test) method.

$$H0 : \beta i = 0$$

$$Ha :\ \beta i \neq 0\ (i = 1,2,\ldots,5)$$

```
factor(Country)CHINA:factor(Reactor_Type)FBR                    0.02886 *
factor(Country)CHINA:Average_Concrete_Price                    0.00243 **
factor(Country)RUSSIA:Average_Concrete_Price                   0.00780 **

Residual standard error: 1.334 on 176 degrees of freedom
Multiple R-squared:  0.7802,    Adjusted R-squared:  0.6753
F-statistic: 7.437 on 84 and 176 DF,  p-value: < 0.00000000000000022
```

Figure : 15 *The result of our Interaction model*

Our interact model has some of the interaction terms which are significant.

Now we shall see the summary of our intercept model by removing all the insignificant predictor variables.

$$H0 : \beta i = 0$$

$$Ha : \ \beta i \neq 0 \ (i = 1,2,\ldots,5)$$

Now we have to manually add all the significant predictor variables to the "cleaned" interaction model. These variables were found to be interaction with each other: -

1) $Country \ * Reactor\ Type$
2) $Country * Average\ Concrete\ Price$

The new interaction model is the following: -

$$interact.new.nuclear.regression.model.reduced.cleaned$$
$$< -lm(Cost_Billion_Dollars \sim factor(Country)$$
$$+ factor(Reactor_Type) + Gross_Capacity_MWe + Duration$$
$$+ Average_Concrete_Price + factor(Country) * factor(Reactor_Type)$$
$$+ factor(Country) * Average_Concrete_Price \ , data \ = \ nuclear.df)$$

```
factor(Country)IRAN, ISLAMIC REPUBLIC OF                          **
Gross_Capacity_MWe                                               ***
factor(Country)CHINA:factor(Reactor_Type)HTGR                    ***
factor(Country)UNITED STATES OF AMERICA:Average_Concrete_Price    *

Residual standard error: 1.286 on 207 degrees of freedom
Multiple R-squared:  0.7598,    Adjusted R-squared:  0.6983
F-statistic: 12.36 on 53 and 207 DF,  p-value: < 0.00000000000000022
```

Figure : 16 *The result of our "cleaned" Interaction model*

As we can see from above t-test, we have discovered that $Country *$ $Average\ Concrete\ Price$ and $Country \ * Reactor\ Type$ independent variables are also quite statistically significant since their p-value < 0.05. Therefore we have to reject the NULL Hypothesis for those variables.

Also note that the Adjusted R-Squared of our **interaction model** is **0.6983** as opposed to **0.4784** for our purely **additive model.** Therefore our interaction model is superior and we must accept this model. We can observe interaction effect.

Interaction Model Interpretation

Final Equation formed: -

$$Cost_Billion_Dollars = -2.0353231 + 9.5331245 * Country(IRAN) + 0.0013776 * Gross_Capacity_MWe - 0.0085935 * Duration + 0.0214370 * Average_Concrete_Price + 14.1141829 * Country(CHINA) * Reactor_Type(HTGR) + 0.0436586 * Country(USA) * Average_Concrete_Price$$

4 Sub-models can be formed from the above equation. It depends upon the value of Country and Reactor Type predictor variable; which sub model shall be used in the end.

Defining and interpreting each sub model: -

1) Country = Every country except for Iran, China and USA, Reactor Type = Can be any reactor type: -

$$Cost_Billion_Dollars = -2.0353231 + 0.0013776 * Gross_Capacity_MWe - 0.0085935 * Duration + 0.0214370 * Average_Concrete_Price$$

Interpretation : -

- For every increase in Gross Capacity MWe by 1 MWe, there is an increase of 0.0013776 Billion $ of the reactor cost; keeping all other variables constant.
- For every 1 year increase in the Duration, there is an decrease of 0.0085935 Billion $ of the reactor cost; keeping all other variables constant.
- For every increase in Average Concrete Price by 1 unit, there is an increase of 0.0214370 Billion $ of the reactor cost; keeping all other variables constant.

2) Country = IRAN, Reactor Type = Can be any reactor type: -

$$Cost_Billion_Dollars = -2.0353231 + 9.5331245 * Country(IRAN) + 0.0013776 * Gross_Capacity_MWe - 0.0085935 * Duration + 0.0214370 * Average_Concrete_Price$$

Interpretation : -

- For every increase in Gross Capacity MWe by 1 MWe, there is an increase of 0.0013776 Billion $ of the reactor cost; keeping all other variables constant.

- For every 1 year increase in the Duration, there is an decrease of 0.0085935 Billion $ of the reactor cost; keeping all other variables constant.
- For every increase in Average Concrete Price by 1 unit, there is an increase of 0.0214370 Billion $ of the reactor cost; keeping all other variables constant.
- Whenever a nuclear reactor is made in IRAN, the reactor cost 9.5331245 Billion $ more.

3) Country = USA, Reactor Type = Can be any reactor type: -

$$Cost_Billion_Dollars = -2.0353231 + 0.0013776 * Gross_Capacity_MWe - 0.0085935 * Duration + (0.0214370 + 0.0436586 * Country(USA)) * Average_Concrete_Price.$$

Since Country(USA) == TRUE, Therefore value of Country(USA) = 1. Hence: -

$$Cost_Billion_Dollars = -2.0353231 + 0.0013776 * Gross_Capacity_MWe - 0.0085935 * Duration + 0.0650956 * Average_Concrete_Price$$

Interpretation : -

- For every increase in Gross Capacity MWe by 1 MWe, there is an increase of 0.0013776 Billion $ of the reactor cost; keeping all other variables constant.
- For every 1 year increase in the Duration, there is an decrease of 0.0085935 Billion $ of the reactor cost; keeping all other variables constant.
- For every increase in Average Concrete Price by 1 unit, there is an increase of 0.0650956 Billion $ of the reactor cost; keeping all other variables constant.

4) Country = CHINA, Reactor Type = HTGR : -

$$Cost_Billion_Dollars = -2.0353231 + 0.0013776 * Gross_Capacity_MWe - 0.0085935 * Duration + 0.0214370 * Average_Concrete_Price + 14.1141829 * Country(CHINA) * Reactor_Type(HTGR)$$

Interpretation : -

- For every increase in Gross Capacity MWe by 1 MWe, there is an increase of 0.0013776 Billion $ of the reactor cost; keeping all other variables constant.

- For every 1 year increase in the Duration, there is an decrease of 0.0085935 Billion $ of the reactor cost; keeping all other variables constant.
- For every increase in Average Concrete Price by 1 unit, there is an increase of 0.0214370 Billion $ of the reactor cost; keeping all other variables constant.
- When CHINA is making a nuclear reactor of type HTGR, then it costs them 14.1141829 Billion dollars more to construct that reactor.

Higher Order Model

We shall now try to make a higher order model by keeping our interaction model as our foundation. We shall now use GGPairs for our higher order model.

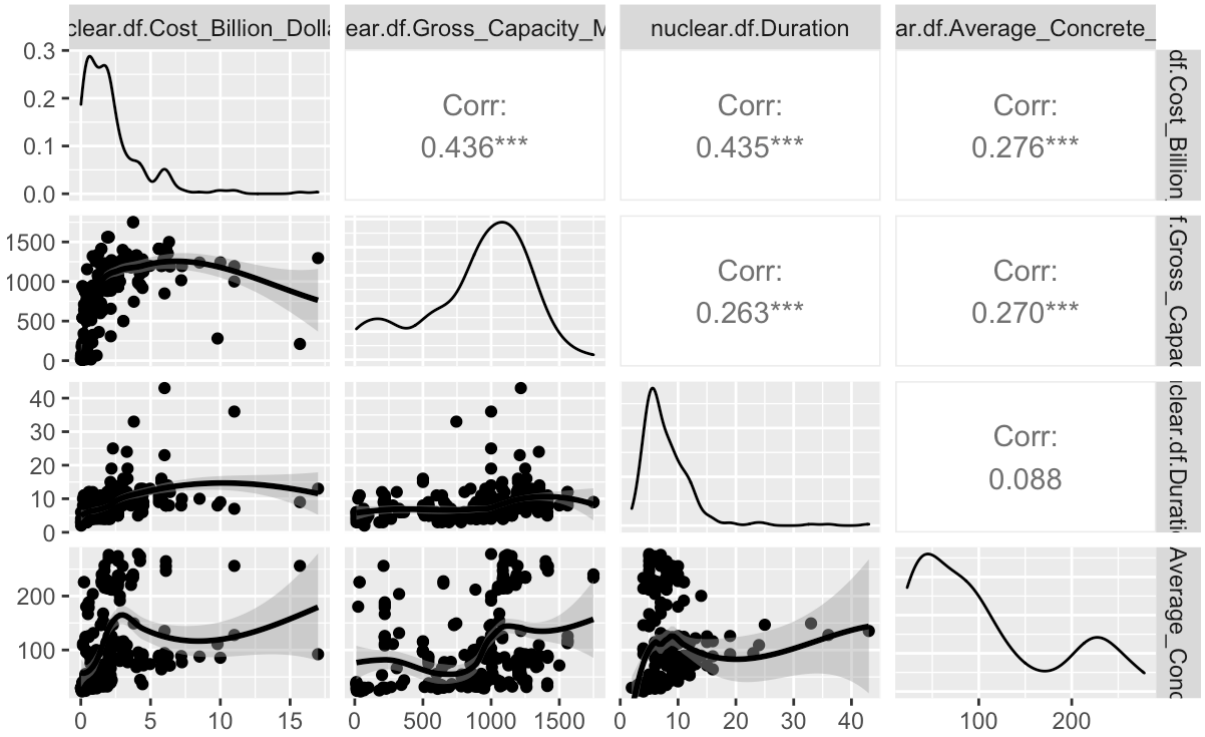

Figure : 17 *The result of our GG Pairs*

In our interact model, we should start by adding Higher orders of

Gross Capacity MWe because it has the highest correlation with response variable Cost .

(0.436)

```
Gross_Capacity_MWe                                          0.96404
I(Gross_Capacity_MWe^2)                                     0.17259
```

Figure : 18 *Higher Order Model by Adding Gross Capacity MWe^2*

Gross Capacity MWe^2 variable is not significant. We tried the same with the variable Duration and Average Concrete Price. Not one variable was significant enough. Therefore our Interaction Model is the best model.

Linearity Assumption Test

We shall now be using our Interaction Model for all the tests.

Plotting the residuals vs predicted value Ŷ plot.

Residual vs fitted data plot.

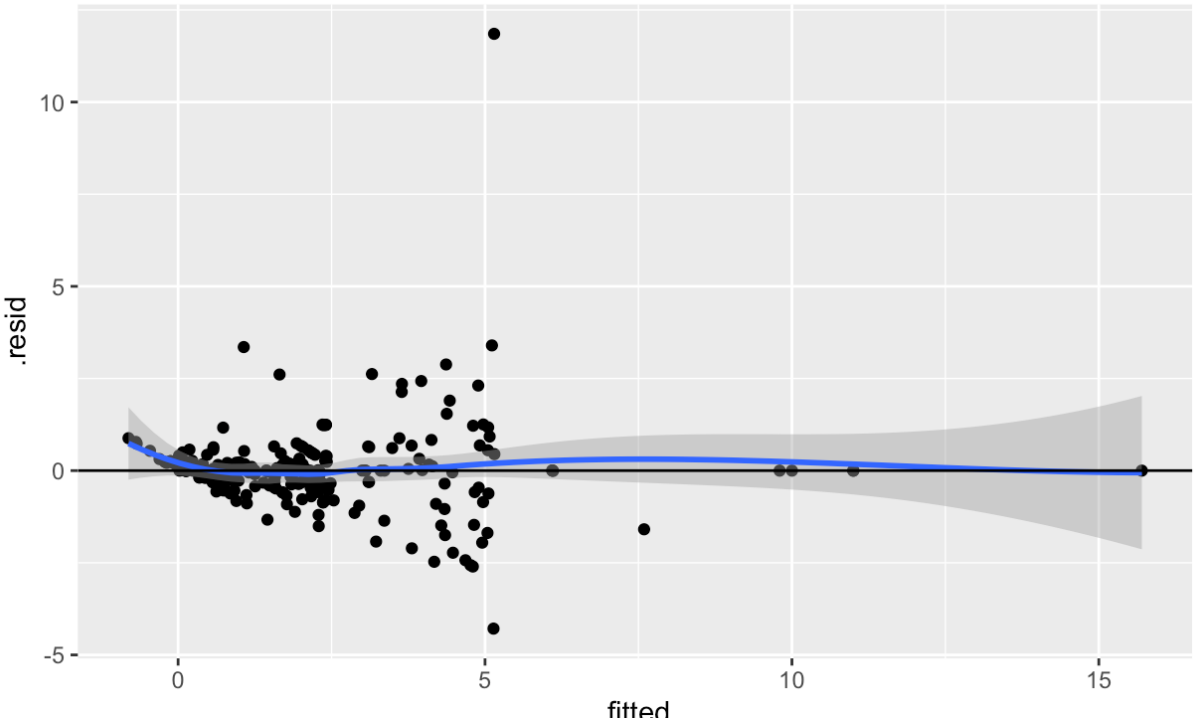

Figure : 19 *Residual V/S Fitted Plot*

Residual plot looks almost perfectly linear.

Independence Assumption Test

Independence Assumption Test usually fails when the data for both dependent and independent variables are observed sequentially over a period of time-called time-series data. Since our dataset in NOT time-series data, we pass the Independence Assumption Test.

Equal Variance Assumption

Checking for Homoscedasticity Assumption. Using Bursch Pagan in order to check for heteroscedasticity.

$$H0: heteroscedasticity\ is\ not\ present\ (homoscedasticity)$$

$$Ha: heteroscedasticity\ is\ present$$

```
        studentized Breusch-Pagan test

data:  nuclear.best.model
BP = 19.912, df = 53, p-value = 1
```

Figure : 20 *BP Test Result*

As we can see from the BP Test, the p-value > 0.05. Therefore H0 cannot be rejected. This means that in our best fit model, the error terms have a constant variance i.e. homoscedasticity.

Plotting a scale location plot for visualisation.

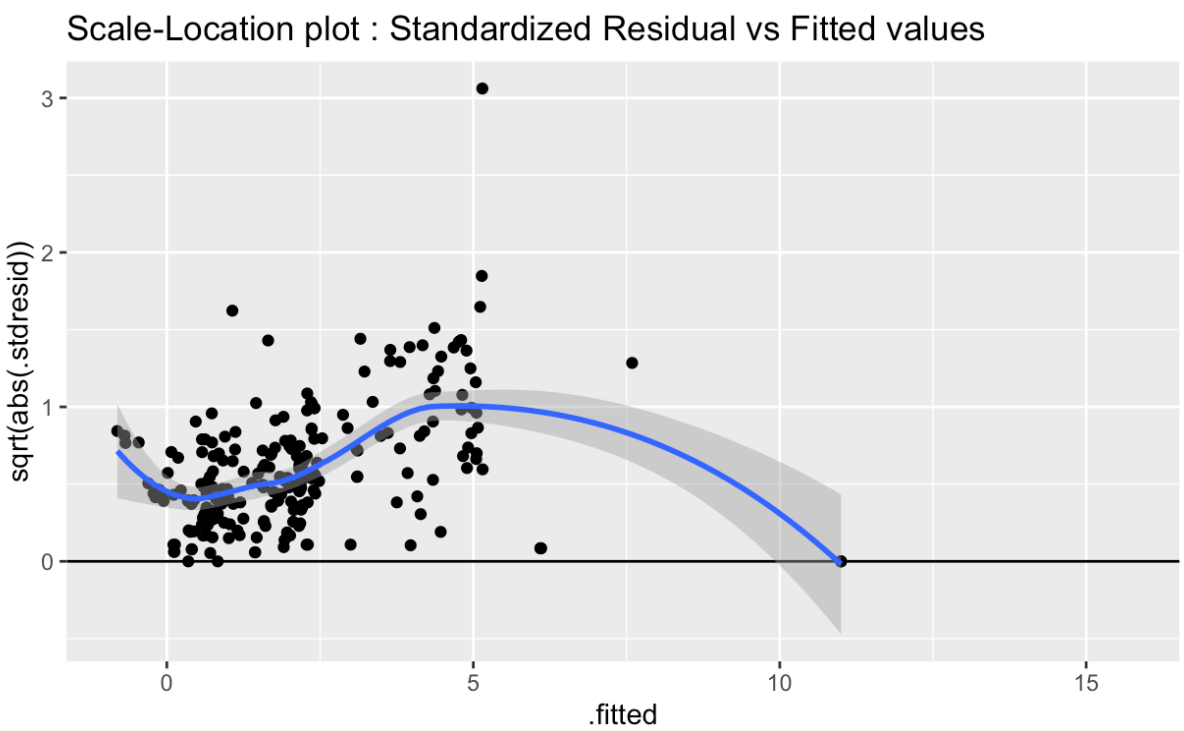

Figure : 21 *Scale Location Plot*

Normality Assumption

Checking for Normality Assumption using Shapiro - Wilk test.

$$H0: the\ sample\ data\ are\ significantly\ normally\ distributed$$

$$Ha: the\ sample\ data\ are\ not\ significantly\ normally\ distributed$$

```
        Shapiro-Wilk normality test

data:  residuals(nuclear.best.model)
W = 0.67977, p-value < 0.00000000000000022
```

Figure : 22 *Shapiro-Wilk Test Result*

From the Shapiro-Wilk normality test, we can see that the p-value < 0.05. Therefore H0 has to be rejected. This means that the sample data are not significantly normally distributed. This is a problem.

Plotting a Normal Q-Q plot for visualisation.

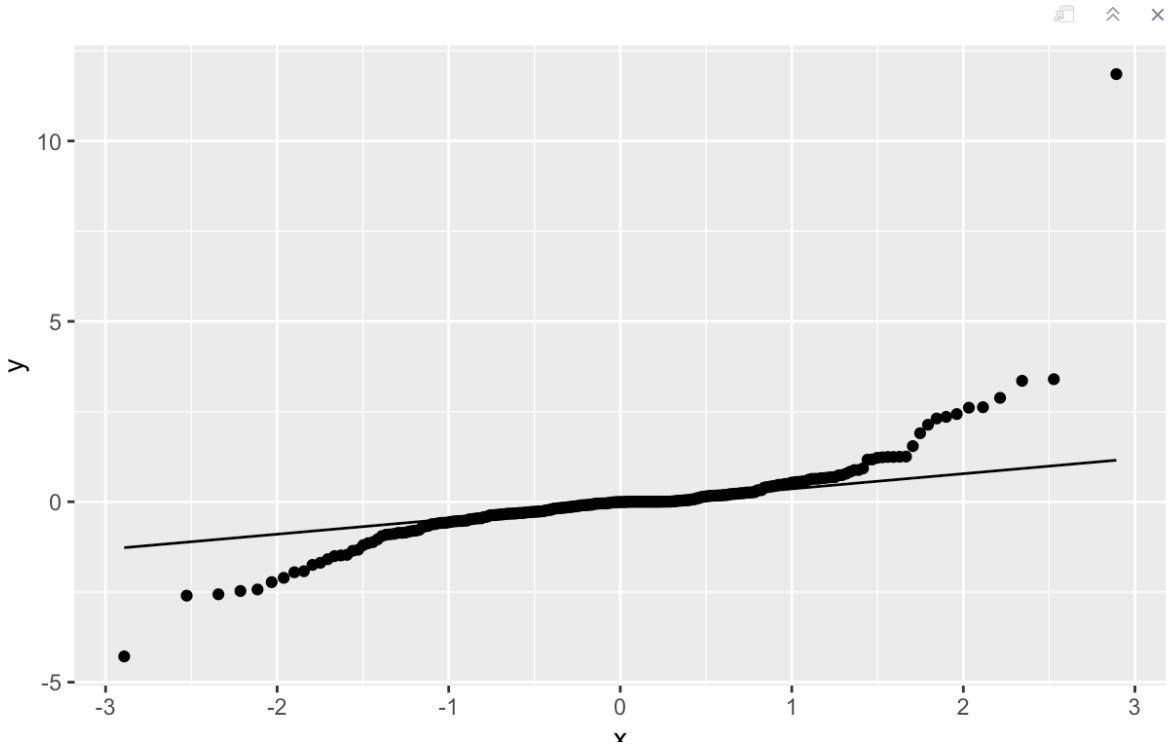

Figure : 23 *Normal Q-Q Plot*

An S-shaped pattern of deviations indicates that the residuals have excessive kurtosis, i.e., there are either too many or two few large errors in both directions.

Checking for Outliers

Detecting outliers by using Cook's distance measure (using cooks.distance()>0.5) and Residual vs Leverage plot.

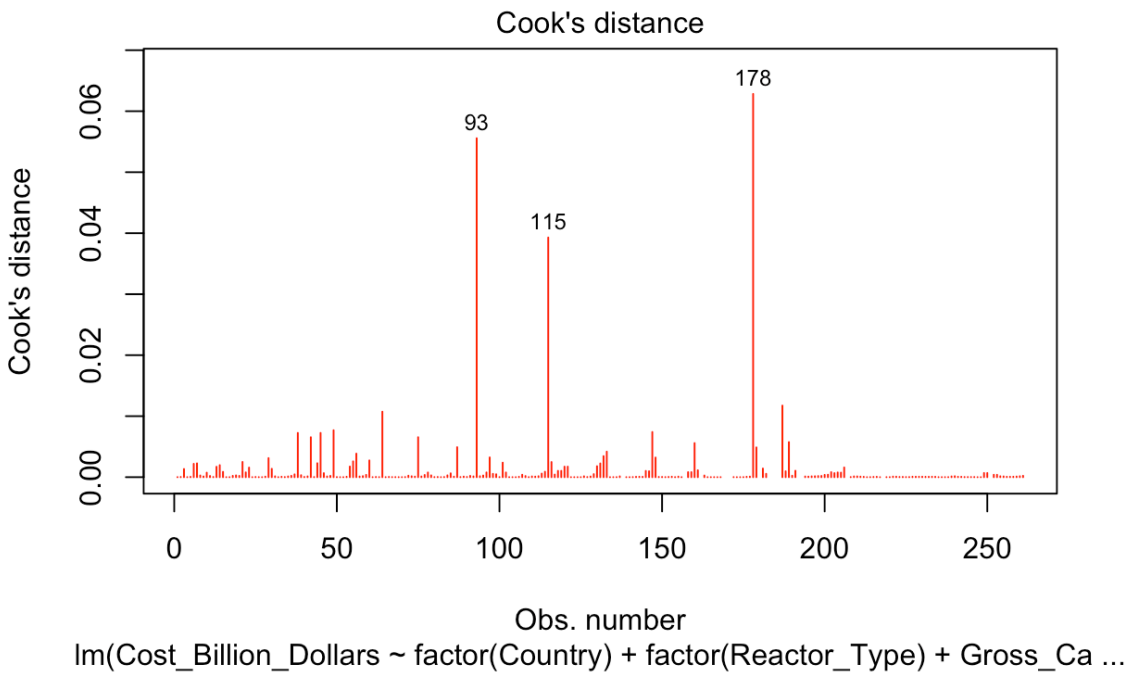

Figure : 24 *Cook's Distance*

There is not a single data point whose cook's distance is > 0.5. Hence according to Cook, no outlier detected.

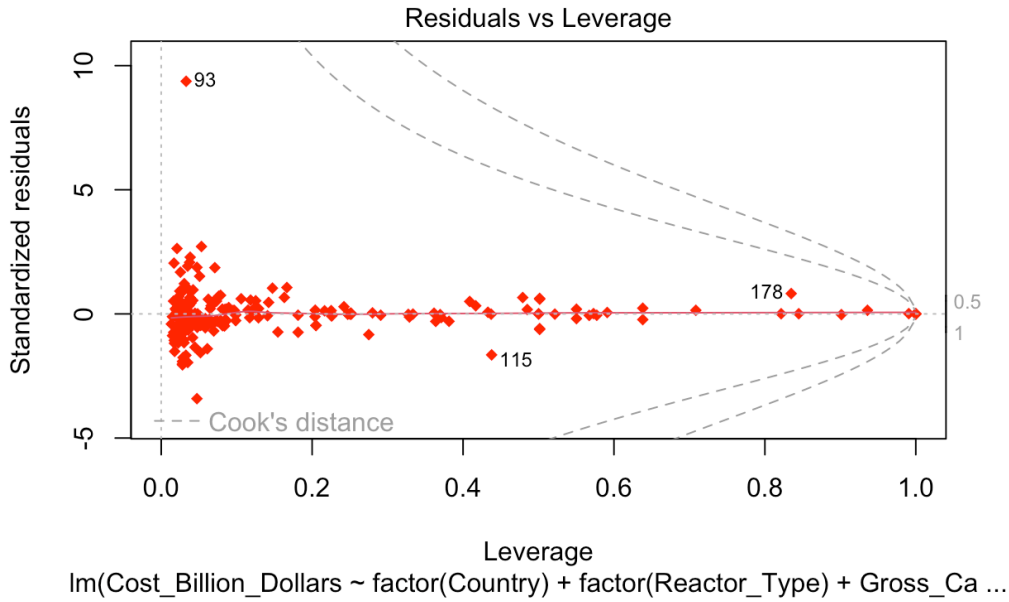

Figure : 25 *Cook's Distance*

Again no outlier found.

Box Cox Transformation

In order to correct the Normality Assumption, we shall try Box-Cox Transformation.

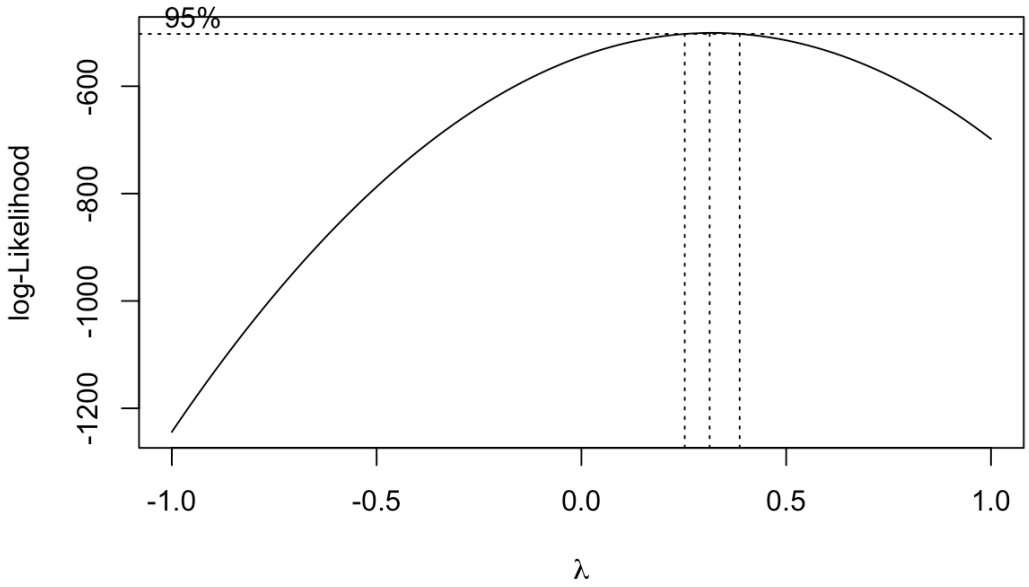

Figure : 26 *Box Cox Transformation*

Extracting best lambda = 0.3131313.
Our New Equation is therefore: -

$$box.cox.nuclear.best.model < -lm((((Cost_Billion_Dollars\text{^}0.3131313) - 1)/0.3131313) \sim factor(Country) + factor(Reactor_Type) + Gross_Capacity_MWe + Duration + Average_Concrete_Price + factor(Country) * factor(Reactor_Type) + factor(Country) * Average_Concrete_Price, data = nuclear.df)$$

Our model without Box-Cox transformation had the R^2Adj value of **0.6983**, whereas after transformation it went up to **0.8019**. This is a huge change in positive direction. Box-Cox made our model better.

Performing a diagnostics analysis for the Box-Cox model.

Now we would perform Equal Variance Assumption and Normality Assumption.

Conducting a test for heteroscedasticity (non-constant variance) and plotting a residual plot. Using Bursch Pagan in order to check for heteroscedasticity.

$$H0: heteroscedasticity\ is\ not\ present\ (homoscedasticity)$$

$$Ha: heteroscedasticity\ is\ present$$

```
        studentized Breusch-Pagan test

data:  box.cox.nuclear.best.model
BP = 48.411, df = 53, p-value = 0.6532
```

Figure : 27 *BP Test Result*

As we can see from the BP Test, the p-value > 0.05. Therefore H0 cannot be rejected. This means that in our box cox fit model, the error terms have a constant variance i.e. homoscedasticity. Do note that the p-value decreased after the transformation.

Plotting a scale location plot for visualisation.

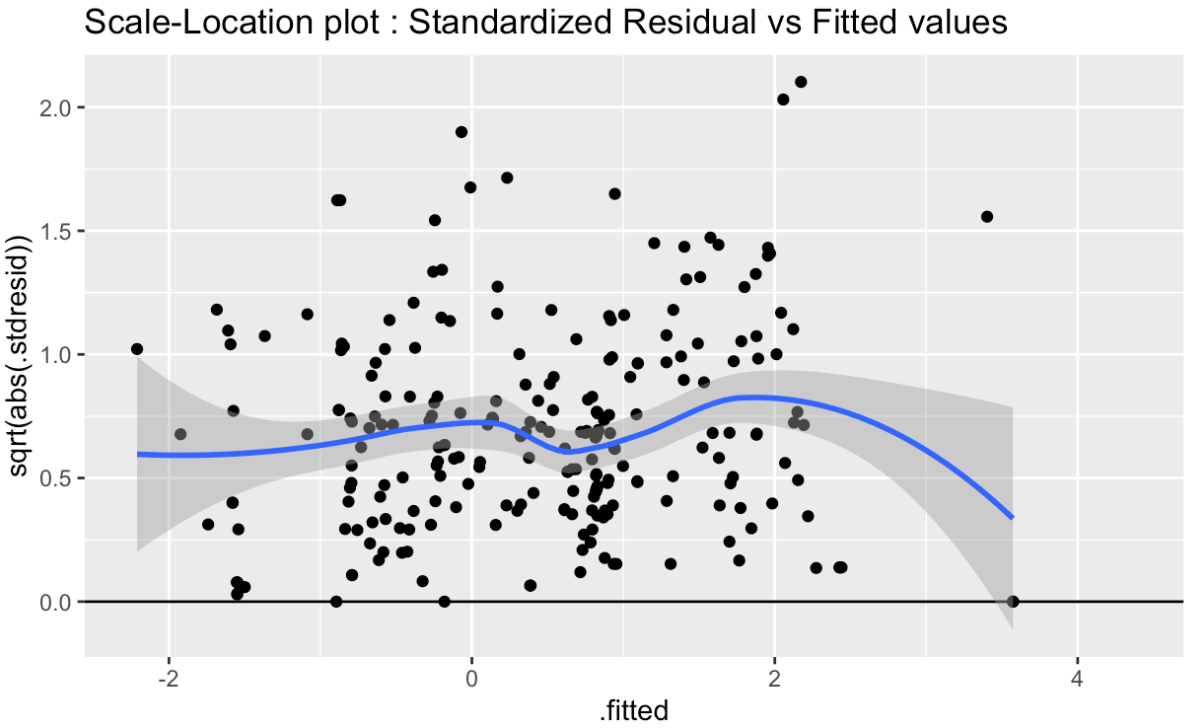

Figure : 28 *Scale Location Plot*

Checking for Normality Assumption using Shapiro - Wilk test.

$$H0: the\ sample\ data\ are\ significantly\ normally\ distributed$$

$$Ha: the\ sample\ data\ are\ not\ significantly\ normally\ distributed$$

```
Shapiro-Wilk normality test

data:  residuals(box.cox.nuclear.best.model)
W = 0.91051, p-value = 0.00000000002259
```

Figure : 29 *Shapiro-Wilk Test Result*

From the Shapiro-Wilk normality test, we can see that the p-value < 0.05. Therefore H0 has to be rejected. This means that the sample data is still not significantly normally distributed. Do note that the p-value of Shapiro-Wilk normality test has improved sharply after the transformation, but it is still not enough.

Plotting a Normal Q-Q plot for visualisation.

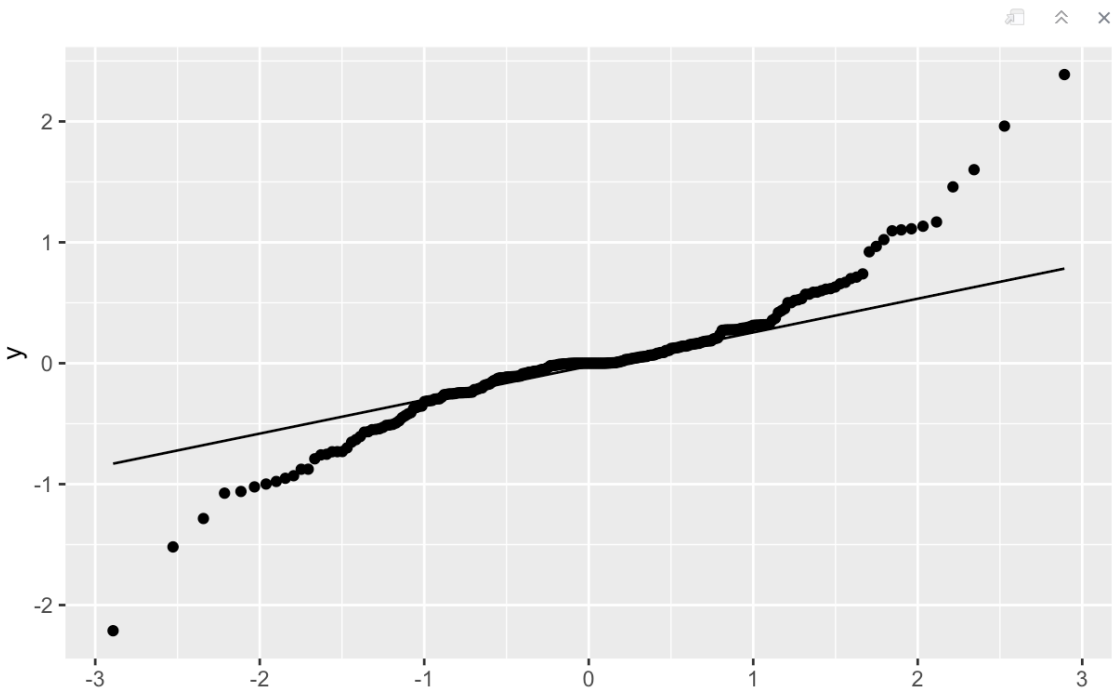

Figure : 30 *Normal Q-Q Plot*

Discussion

Summary

We started with 10 predictor variables: -

1) *Country*
2) *Reactor Type*
3) *Gross Capacity (MWe)*
4) *Thermal Capacity (MWt)*
5) *Efficiency Percentage*
6) *Starting Year*
7) *Duration (in years)*
8) *Average Inflation Percentage*
9) *Average Concrete Price*
10) *Average Steel Price*

3 predictor variables got eliminated in Multicollinearity testing: -

1) *Thermal Capacity (MWt)*
2) *Average Steel Price*
3) *Starting Year*

Thermal Capacity (MWt) got eliminated because of high correlation with *Gross Capacity (MWe).* This makes sense because the more heat your reactor is designed to generate, the more electricity you can produce from it.

Average Steel Price got eliminated because of high correlation with *Average Concrete Price*. This makes sense because the prices of Concrete and Steel has been rising through the years in the same proportions.

Starting Year Price got eliminated because of high correlation with *Average Concrete Price*. This makes sense because the prices of Concrete strongly depends on the Starting Year. The prices were cheaper in earlier years when compared with the present.

We are now left with 7 variables for our additive model: -

1) *Country*
2) *Reactor Type*
3) *Gross Capacity (MWe)*
4) *Efficiency Percentage*
5) *Duration (in years)*
6) *Average Inflation Percentage*
7) *Average Concrete Price*

After Individual T-test and other procedures, we found that the following variables were insignificant: -

1) *Efficiency Percentage*
2) *Average Inflation Percentage*

Efficiency Percentage not being significant was surprising to us. We thought that efficient turbines must be very expensive and therefore would increase the construction cost of the reactor. However this was not the case. This means that cost of Turbine does not matter a lot in the overall cost.
Average Inflation Percentage also does not matter when a reactor is getting constructed.

We are now left with 5 variables for our interaction model: -

1) *Country*
2) *Reactor Type*
3) *Gross Capacity (MWe)*
4) *Duration (in years)*
5) *Average Concrete Price*

After making our Interaction Model, we found the following variables to be significant: -

1) *Country*
2) *Reactor Type*
3) *Gross Capacity (MWe)*
4) *Duration (in years)*
5) *Average Concrete Price*
6) *Country * Reactor Type*
7) *Country * Average Concrete Price*

We could not make Higher Order Model because the terms were not significant.
When it comes to Assumption Testing, out Interaction Model passed all the tests except for Normality Assumption.
Due to this, we did Box-Cox Transformation (Extracting best lambda = 0.3131313)
However, even after this transformation, we were not able to meet the Normality Assumption.

Our model without Box-Cox transformation had the R^2Adj value of **0.6983**, whereas after transformation it went up to **0.8019**. This is a huge change in positive direction. Box-Cox made our model better.

Therefore our final model is: -

$$box.cox.nuclear.best.model < -lm((((Cost_Billion_Dollars\text{^}0.3131313) - 1)/0.3131313) \sim factor(Country) + factor(Reactor_Type) + Gross_Capacity_MWe + Duration + Average_Concrete_Price + factor(Country) * factor(Reactor_Type) + factor(Country) * Average_Concrete_Price, data = nuclear.df)$$

Hence we can at least conclude that when we are constructing a nuclear reactor, the following things affects it's cost a lot: -

1) *Country*
2) *Reactor Type*
3) *Gross Capacity (MWe)*
4) *Duration (in years)*
5) *Average Concrete Price*

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